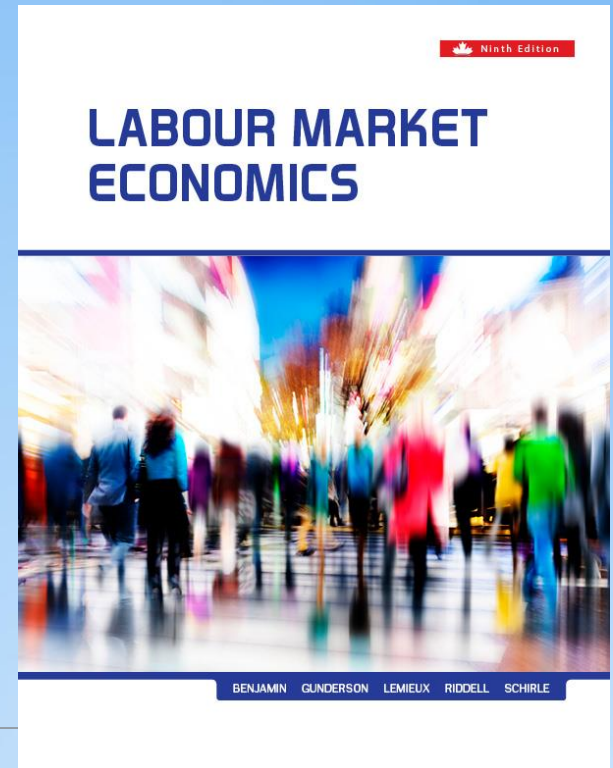


Chapter Two

Labour Supply: Individual Attachment to the Labour Market



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Learning Objectives

1. Define the key elements of labour force measurement - employment, unemployment, labour force participation, and hours worked - and explain how they are measured and reported by Statistics Canada.
2. Illustrate graphically how the income-leisure model reflects the trade-offs that consumers face in deciding whether and how much to work.
3. Distinguish theoretically between the work choices made by individuals and the economic opportunities that they choose from.
4. Explain using diagrams how an increase in the wage rate leads to offsetting income and substitution effects, and how this yields an ambiguous effect of wage changes on labour supply.
5. Interpret the economic and other factors affecting a married woman's decision to work and show how this decision can be captured within the income-leisure (labour supply) model.

Main Questions

- How do we measure labour market attachment? How did labour market attachment evolve over the twentieth century?
- What is the labour force participation decision, and how does it fit into the general theory of labour supply?
- How can we incorporate the possibly different factors that determine men's and women's labour supply decisions?
- Is there any theoretical reason to believe that labour supply increases with the market wage?
- What is the evidence regarding the responsiveness of individual labour supply to changes in the wage?
- Why are workers paid an overtime premium, rather than just a higher wage, in order to induce them to work overtime?

The Theory of Labour Supply

Quantifying Labour Market Attachment

Labour Force (LF)

- Individuals in the eligible population (15 years and older) who participate in labour market activities, either employed or unemployed

Labour Force Participation Rate (LFPR)

- The fraction of the eligible population that participates in the labour force
- $LFPR = LF/POP$

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Unemployment

To be considered unemployed, a person must be in one of the following three categories:

1. Without work but has made specific efforts to find a job within the previous four weeks
2. Waiting to be called back to a job from which he or she has been laid off
3. Waiting to start a new job within four weeks

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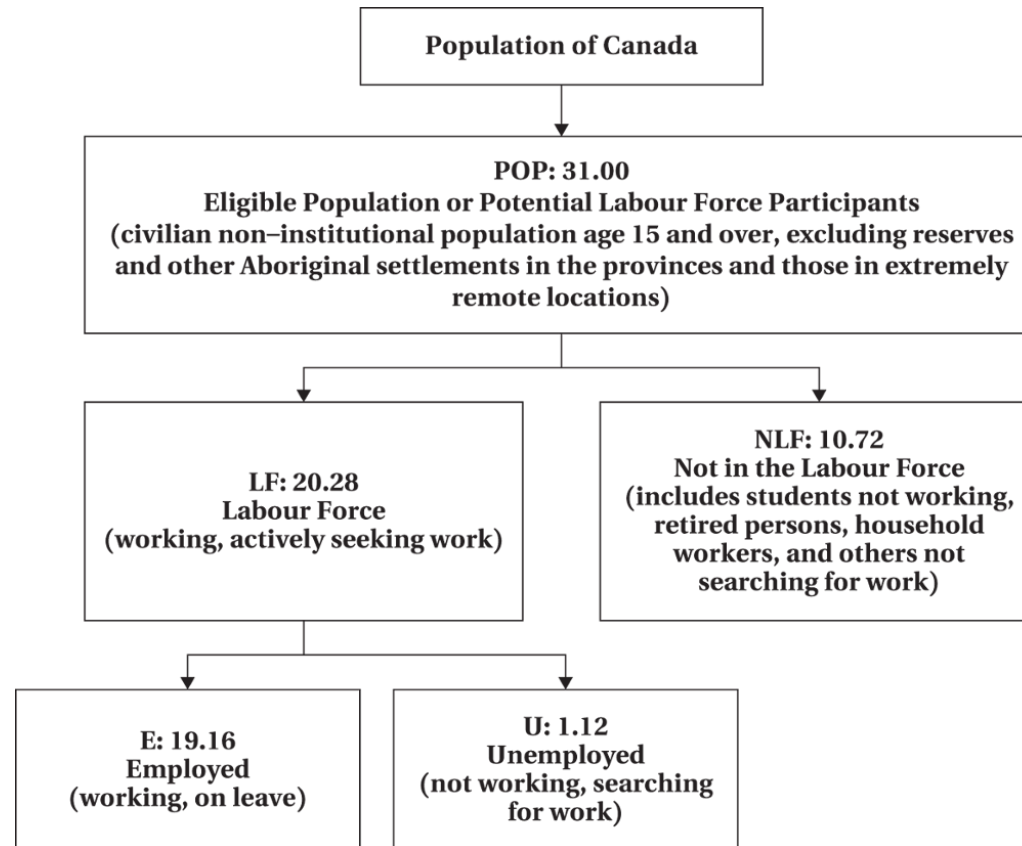


Figure 2.1 Labour Force Concepts (numbers as of January 2020)

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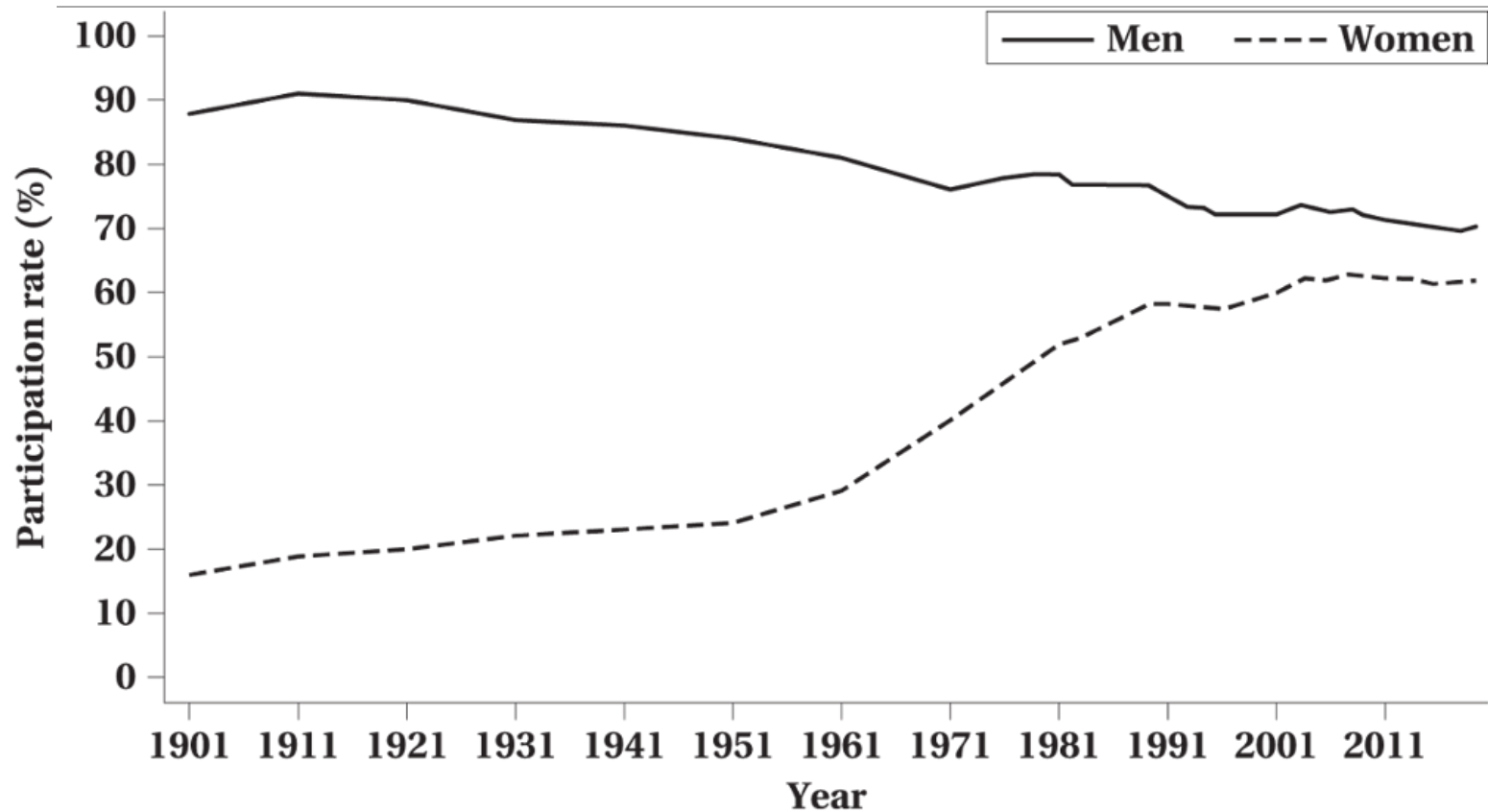


Figure 2.2 Labour Force Participation Rates by Sex, Canada, 1901–2019

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TABLE 2.1 Labour Force Participation Rates by Sex, Selected Countries, 2018

Country	Abbreviation	Overall	Male	Female	Difference
Sweden	SWE	82.9	84.8	81.0	3.7
Vietnam	VNM	82.7	86.4	79.1	7.3
Peru	PER	80.3	88.2	73.3	13.9
Canada	CAN	78.5	81.9	75.1	6.8
United Kingdom	GBR	77.5	82.2	72.8	9.4
Norway	NOR	77.3	79.2	75.2	4.0
Russia	RUS	74.1	79.8	68.9	10.9
Jamaica	JAM	72.4	78.0	66.9	11.1
United States	USA	72.3	77.7	66.8	10.9
France	FRA	71.7	75.8	67.5	8.2
Kenya	KEN	66.9	69.6	64.1	5.5
Italy	ITA	65.3	74.9	55.7	19.2
Mexico	MEX	64.7	82.6	47.1	35.5
Philippines	PHL	62.0	76.2	47.7	35.5
Bangladesh	BGD	61.1	83.9	38.1	45.8
South Africa	ZAF	59.6	66.0	53.4	12.5
Nigeria	NGA	55.2	59.9	50.5	9.4
Egypt	EGY	51.1	76.9	24.7	52.2
Jordan	JOR	41.6	67.4	15.1	52.3

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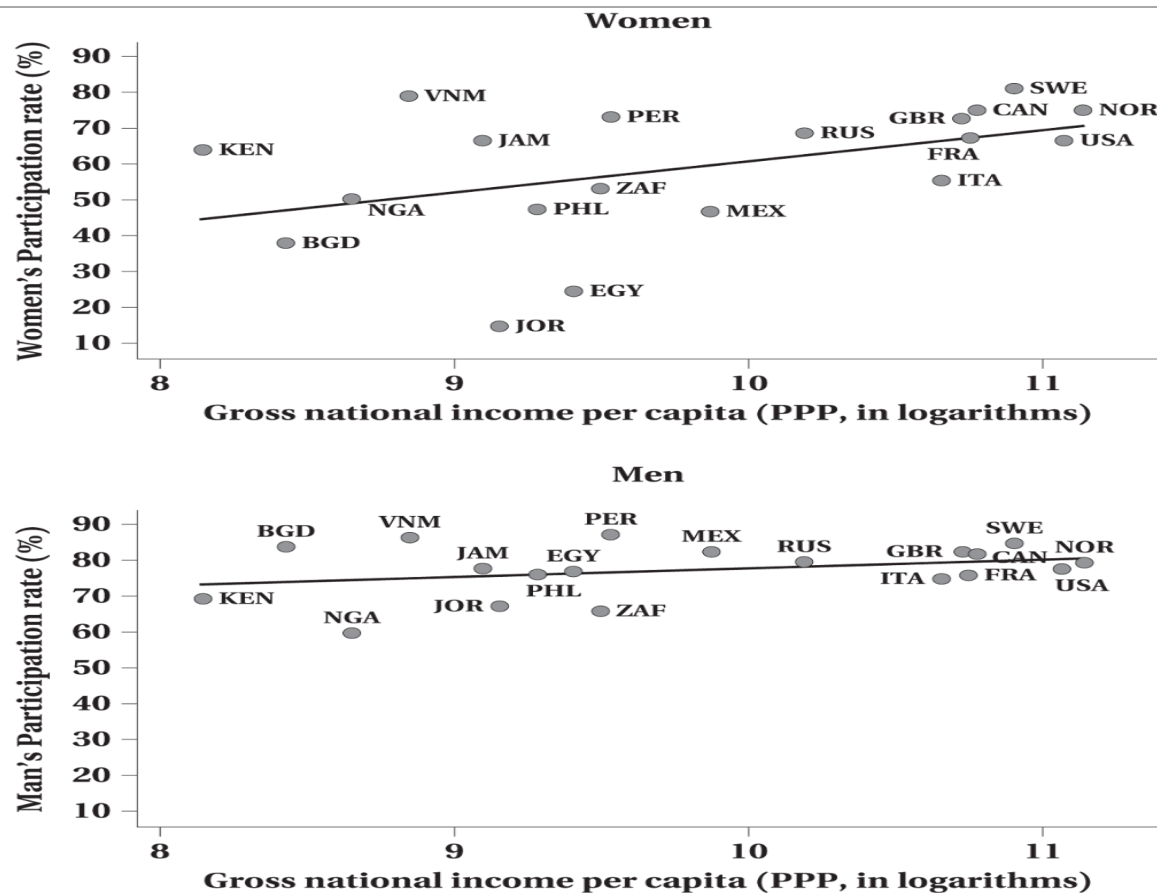


Figure 2.3 Male and Female Participation Rates by Level of Economic Development

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Hours-of-work

- Hours per day, days per week, weeks per year
- It may affect the quantity and the quality of labour supply

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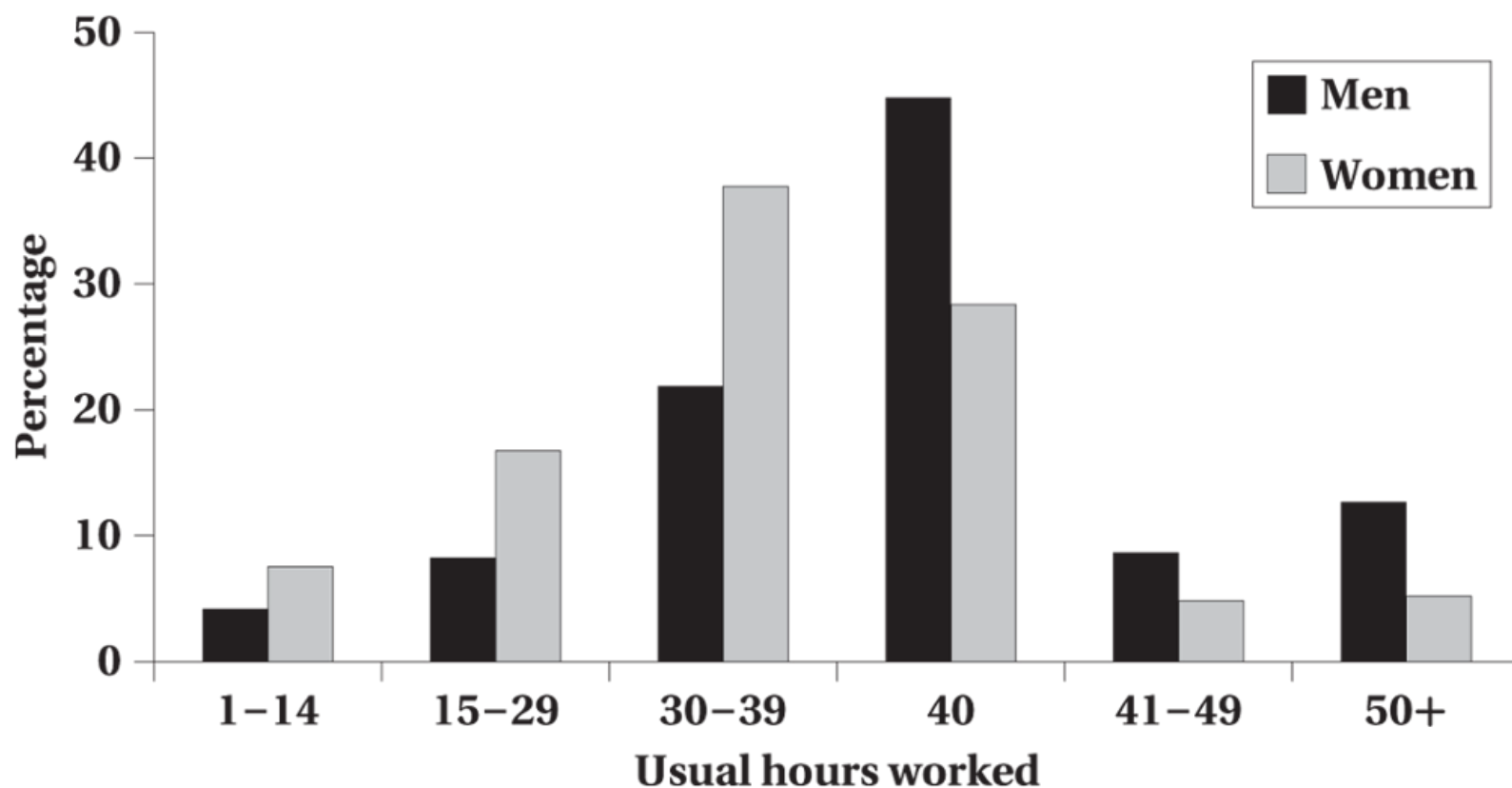


Figure 2.4 Distribution of Hours Worked Per Week by Sex, 2019

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Basic Income-Leisure Model

The choice of hours worked given opportunities and value of non-market time

- Preferences and Constraints
- Individuals choose the feasible outcomes which yield the highest level of satisfaction

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Preferences

Two “goods”

- Consumption
- Leisure

Represented by indifference curves

(A person is indifferent between various combinations of consumption and leisure on an indifference curve)

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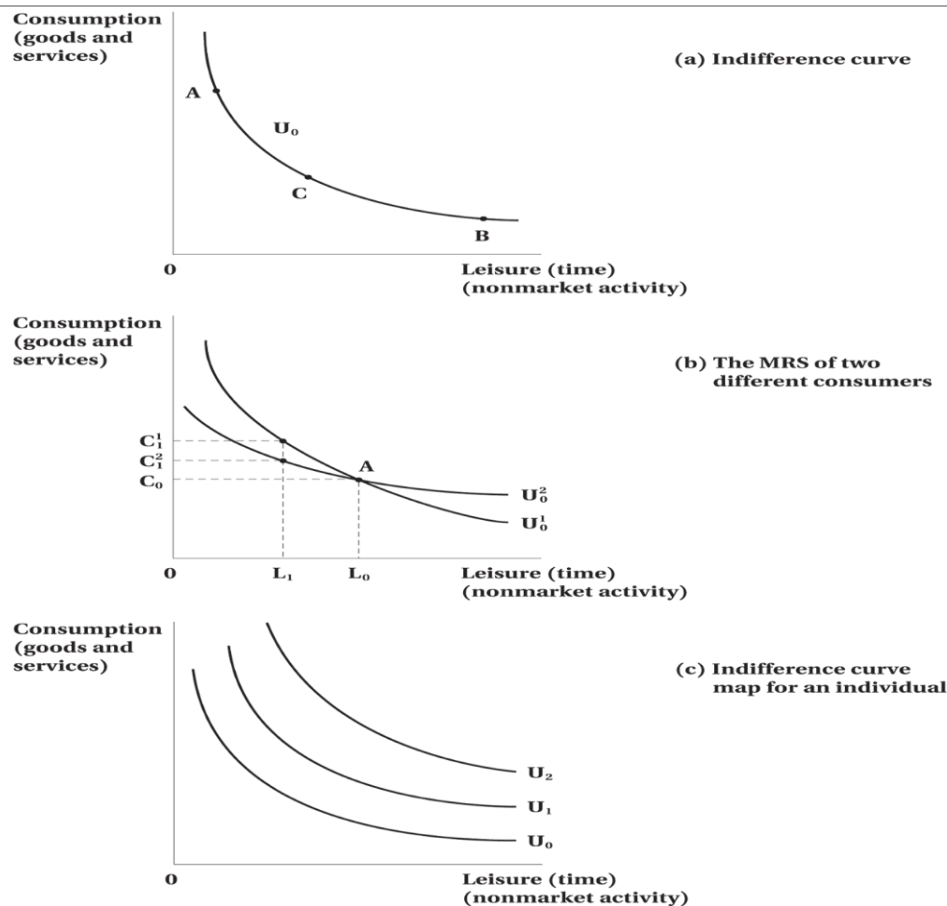


Figure 2.5 Consumer Preferences: Consumption-Leisure Indifference Curves

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Preferences

- We assume that consumers have well-defined preferences over all the conceivable combinations of consumption and leisure.
- This implies that all combinations lie on some indifference curve.
- Consumer preferences can then be represented by an indifference curve.
- Indifference curves plot combinations of consumption and leisure that yield the consumer equal levels of utility.
- The absolute value of the slope of an indifference curve gives the marginal rate of substitution (MRS); that is, the amount of leisure the consumer is willing to accept in exchange for giving up some consumption.

The Theory of Labour Supply

Constrained

Constrained are determined by the economic properties of the market, which, in turn, transform consumption-leisure to income-leisure by setting the price of consumption.

The income and leisure combinations from which the consumer can choose with a *potential income constraint*.

This allows a transformation of our model from representing consumption-leisure tradeoffs to income-leisure tradeoffs.

Alternative phrases used for the potential income constraint include *budget constraint*, *income constraint*, and *full-income constraint*.

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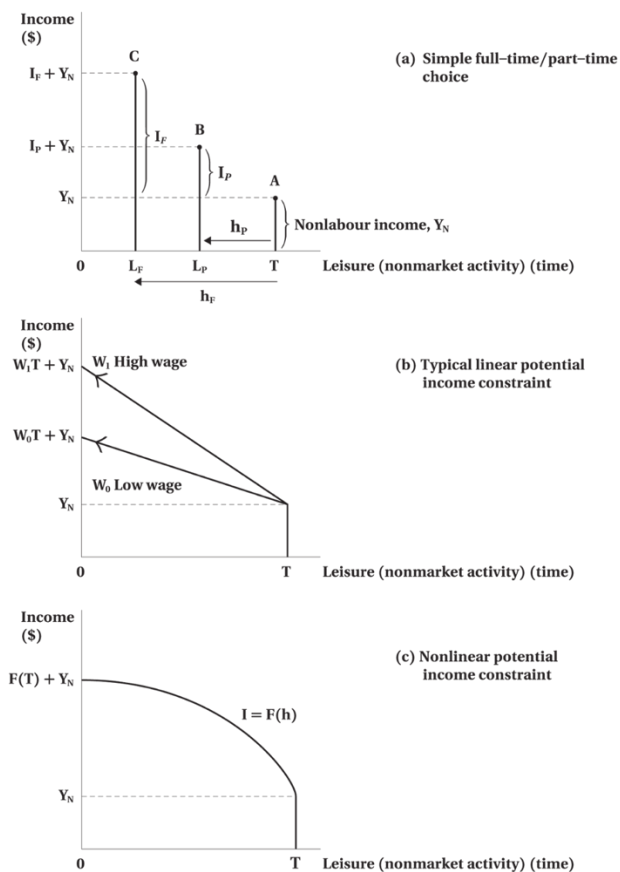


Figure 2.6 Potential Income Constraints: Summarizing Individual Market Opportunities

The Theory of Labour Supply

The Consumer's Optimum

Optimal amount of income and leisure

Utility-maximizing equilibrium

- Highest indifference curve given the income constraint

Compare MRS with the Market Wage Rate

- **MRS:** measures the willingness to exchange leisure for consumption (or income)
- **Market Wage Rate:** measures the ability to exchange leisure for income

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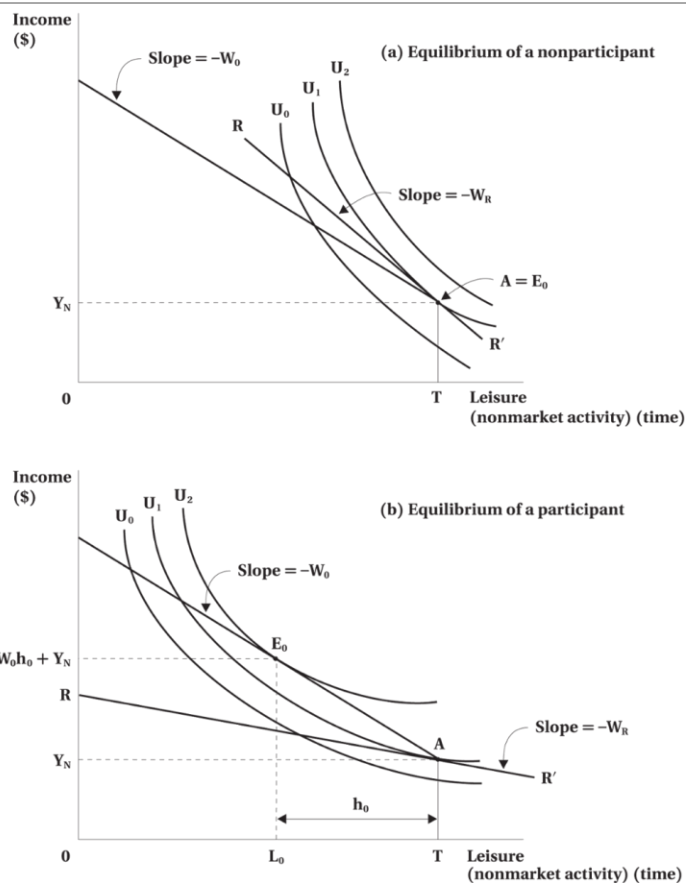


Figure 2.7 The Consumer's Labour Supply Decision

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Worked Example: Working Part-Time at University

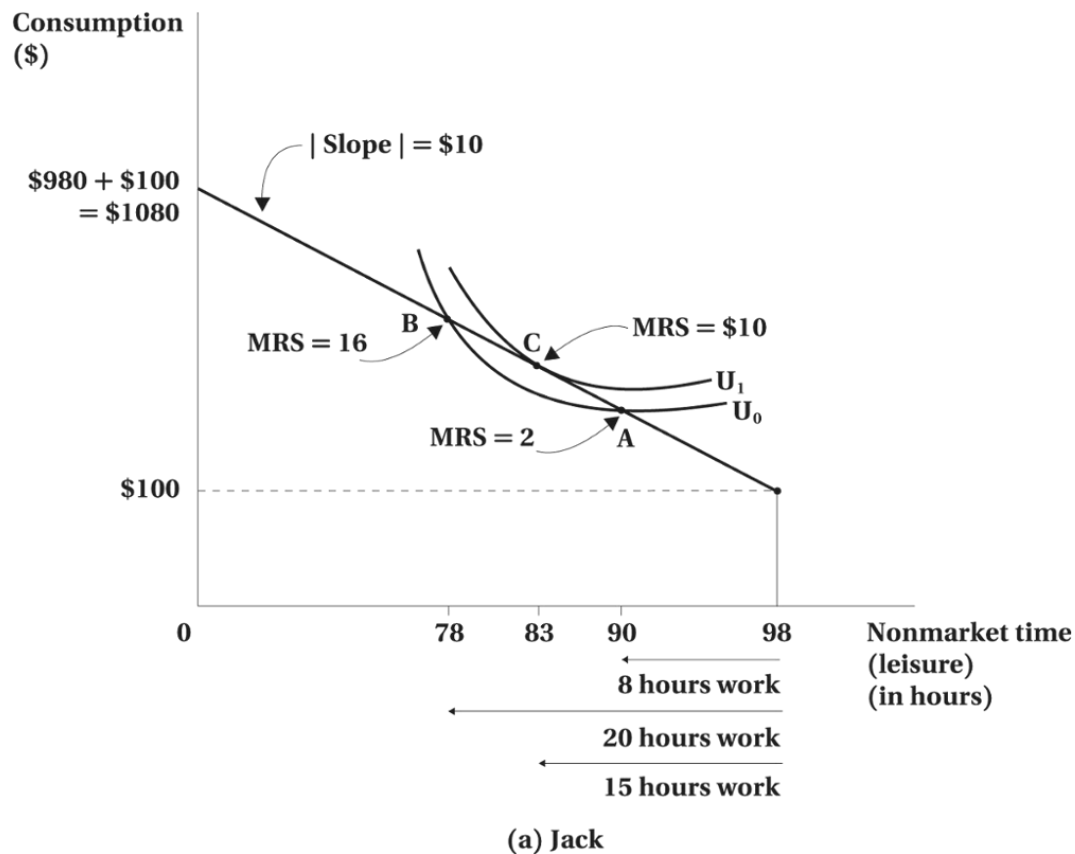
Jack and Jill both attend the same university and are deciding whether to work part-time while they are in school.

Both have identical financial situations: their parents give them a weekly allowance of \$100, and they can work as much as they would like at the local Mega-Mart, at a wage of \$10 per hour.

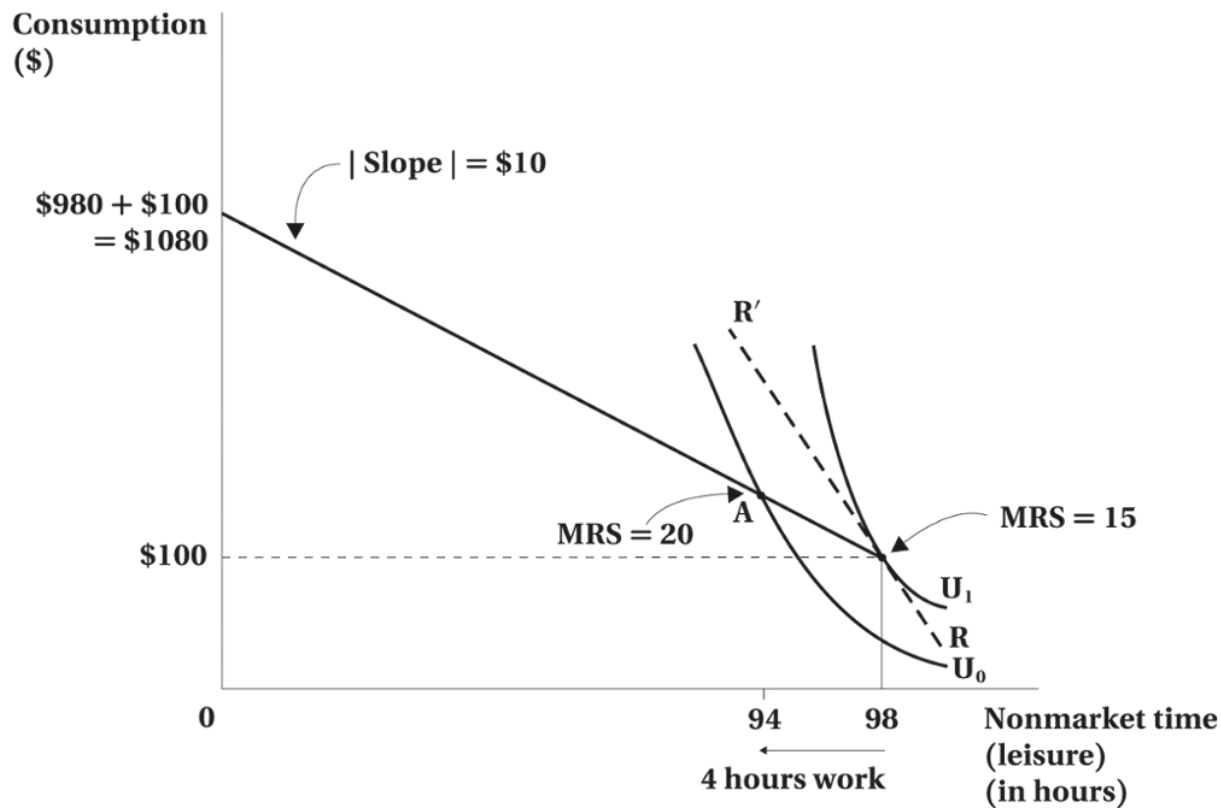
Over a given week, they each have sleep and personal maintenance requirements of 10 hours per day, leaving a total of $7 \times 14 = 98$ hours available for discretionary activities.

Out of this 98 hours, they have to decide how much time to spend working at the Mega-Mart versus time with their studies, partying, socializing with friends, or watching TV.

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(b) Jill

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Comparative Statics

Hours of Work For Participate

Increase in non-labour income results in a parallel outward shift of the budget constraint

- **Leisure, Normal good:**

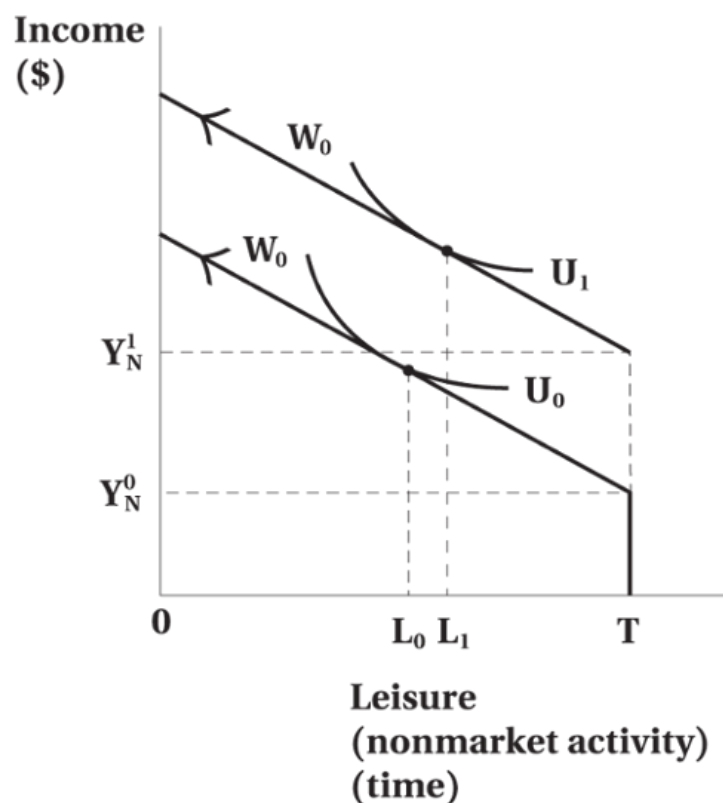
More leisure will be consumed resulting in less work hours

- **Leisure, Inferior good:**

Less leisure will be consumed resulting in more work hours

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(a) Leisure a normal good



(b) Leisure an inferior good

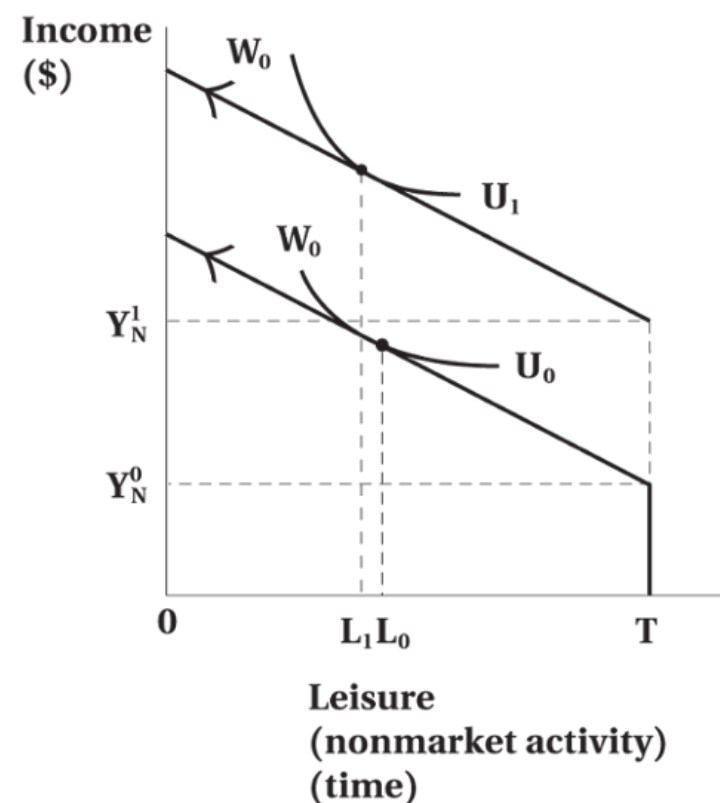


Figure 2.8 The Effect of an Increase in Nonlabour Income on Labour Supply

The Theory of Labour Supply

Change in Wage Rate on Hours of Work

Two effects:

1. **Income effect** The worker has more income to buy more goods including leisure (reduces work hours)
2. **Substitution effect** Individual may work more because the returns are greater substituting away from leisure

The Theory of Labour Supply

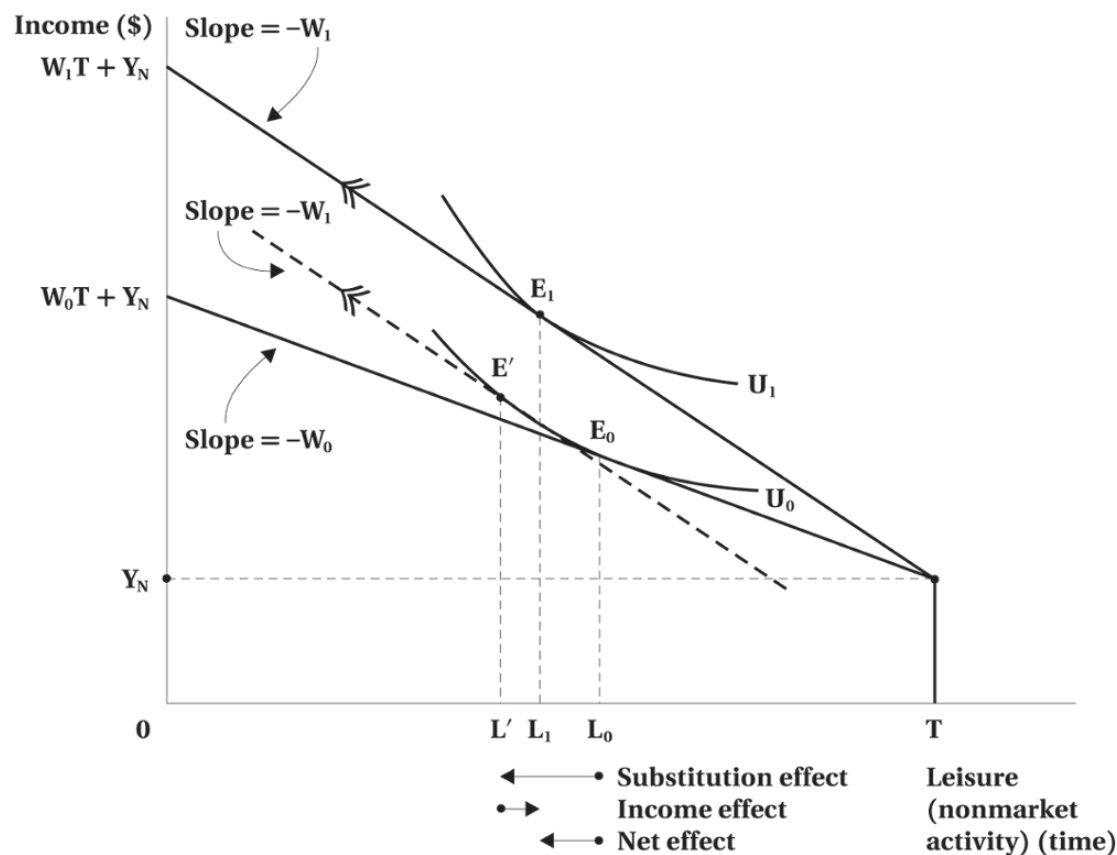


Figure 2.9 Income and Substitution Effects of a Wage Increase

The Theory of Labour Supply

Effect of Wage Increase on Participation

- The net effect depends on both substitution effect and income effect.
- If income effect dominates, hours of work may decline.
- For a non-participant an increase in wages may leave the equilibrium unchanged or induce the individual to participate.

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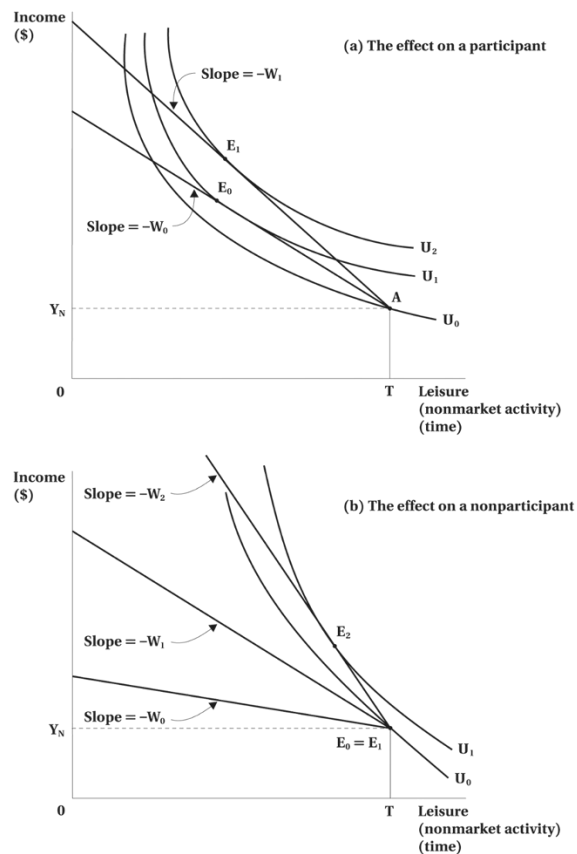


Figure 2.10 The Effect of a Wage Increase on Participation

The Theory of Labour Supply

Deriving the Individual Supply Curve of Labour

If substitution effect dominates,

- Increase in wages leads to increase in labour supplied
- Wages continue to increase until a point where substitution effect and income effect offset each other

Supply curve bends backward when income effect dominates substitution effect

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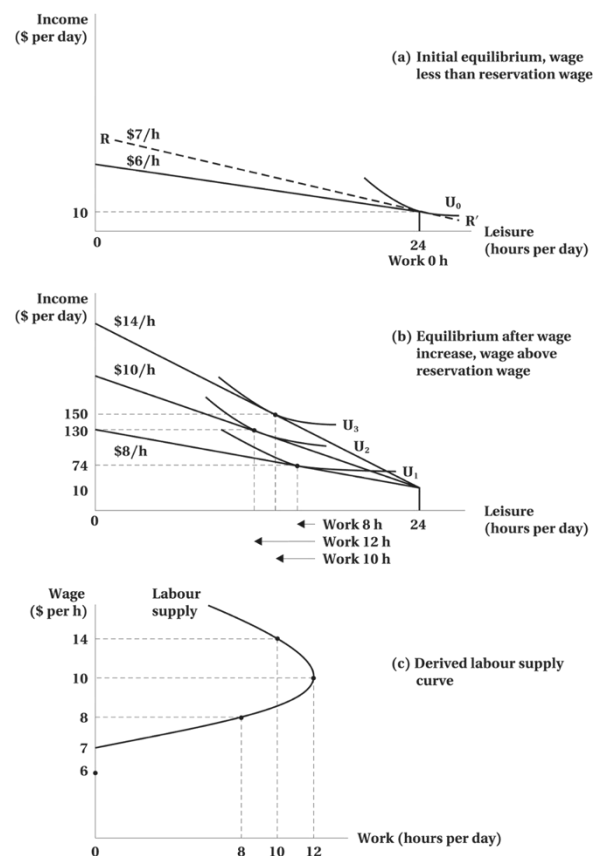


Figure 2.11 Deriving the Individual Supply Curve of Labour

The Theory of Labour Supply

Empirical Evidence

Testing the Model of Labour Supply

Two Research Questions

1. Does labour supply behaviour conform to the predictions of economic theory?
2. How responsive is labour supply to changes in the wage?

TABLE 2.2 Labour Force Participation Rates of Married Women, Canada 2017

	Percentage of Married Women	Participation Rate ^a	Raw Difference from Base Group ^b	Adjusted Difference from Base Group ^c
All Women, Total		75.3		
<i>Age</i>				
20–24	1.1	89.4	N/A	N/A
25–34	18.9	84.4	–5.0	–3.7
35–44	23.9	85.5	–3.9	–3.0
45–54	24.0	84.8	–4.6	–7.3
55–64	23.8	64.8	–24.6	–22.3
65–69	8.4	27.5	–61.9	–52.6
<i>Age of youngest child, under 18 years old, at home</i>				
No children under 18	60.1	70.9	N/A	N/A
0–5 years old	18.4	77.4	6.5	–13.9
6–9 years old	7.7	84.6	13.7	–5.4
10–15 years old	10.1	87.0	16.1	–0.7
16–17 years old	3.7	86.3	15.4	0.2
<i>Education</i>				
Less than high school graduation	8.2	49.6	N/A	N/A
High school graduate or some post-secondary	23.6	67.9	18.3	12.1
Non-university, post-secondary certificate	30.7	77.8	28.2	18.8
University degree	37.6	83.7	34.1	22.4
<i>Spouse's income</i>				
None (or negative)	18.0	49.7	N/A	N/A
\$1–\$9,999	8.7	62.0	12.3	7.7
\$10,000–\$24,999	9.1	78.6	28.9	17.5
\$25,000–\$49,999	18.2	82.3	32.6	18.4
\$50,000–\$74,999	17.1	87.1	37.4	21.0
\$75,000–\$99,999	12.4	86.3	36.6	18.9
\$100,000–\$149,999	10.7	84.2	34.5	17.1
Over \$150,000	5.7	74.1	24.4	7.4

The Theory of Labour Supply

Empirical Evidence

Evidence of the Elasticity of Labour Supply

- Uncompensated elasticity
- Compensated elasticity
- Income elasticity

The Theory of Labour Supply

TABLE 2.3 Estimated Elasticities of Labour Supply

Sex	Uncompensated (Gross, Total) Wage Elasticity of Supply	Compensated (Substitution) Wage Elasticity of Supply	Income Elasticity of Supply
Both sexes	0.0 to 0.2	0.1 to 0.3	−0.1 to 0.0
Men and single women	−0.1 to 0.2	0.1 to 0.3	−0.1 to 0.0
Married women	0.1 to 0.3	0.2 to 0.4	−0.1 to 0.0

Extensions and Applications

Discouraged worker effects

- In periods of high unemployment, people may become discouraged from looking for work and drop out of the labour force

Added worker effects

- In periods of high unemployment, some may enter the labour force to supplement family income that may have deteriorated with the unemployment of other family members.

Extensions and Applications

Hidden Unemployment

- The discouraged worker effect and the problem of hidden unemployment
- “Secondary” or “marginal” workers

Extensions and Applications

Moonlighting, Overtime, Flexible Work Hours

Why do some people moonlight at a second job at a wage less than their market wage on their first job?

Why do some people require an overtime premium to work more?

Extensions and Applications

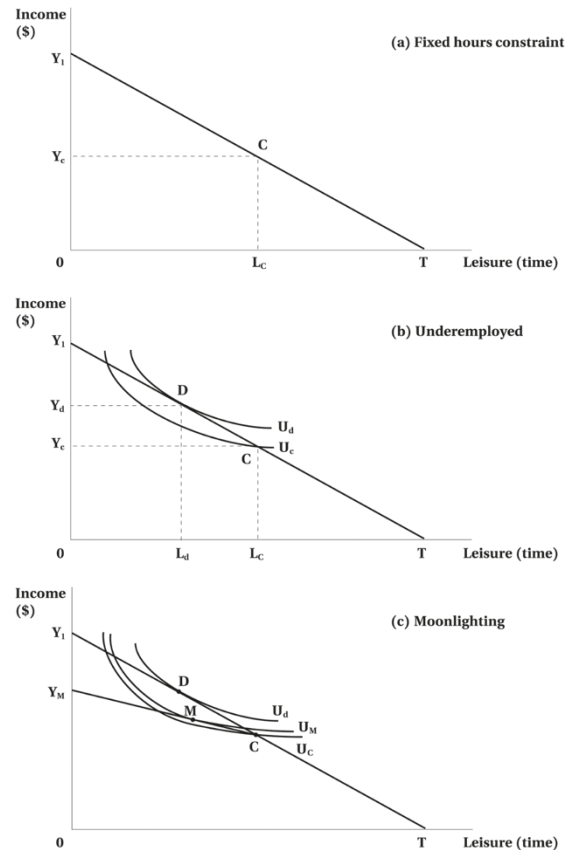


Figure 2.12 Fixed Hours Constraint, Underemployment, and Moonlighting

Extensions and Applications

Overtime and Overemployment

- Workers prefer to work fewer hours at the going wage rate
- Workers are induced to work more hours through an overtime premium

Extensions and Applications

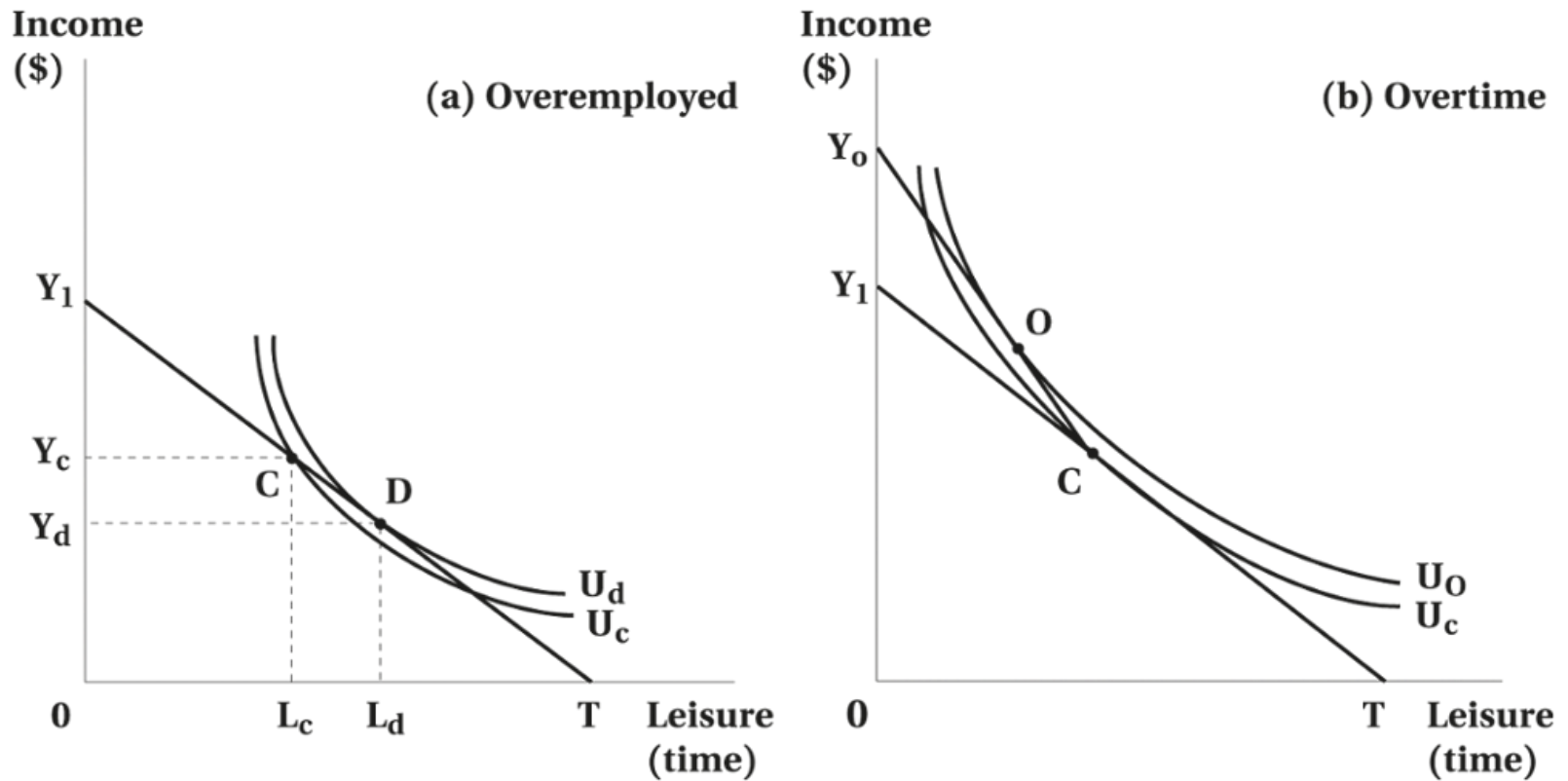


Figure 2.13 Overemployment and Overtime

Extensions and Applications

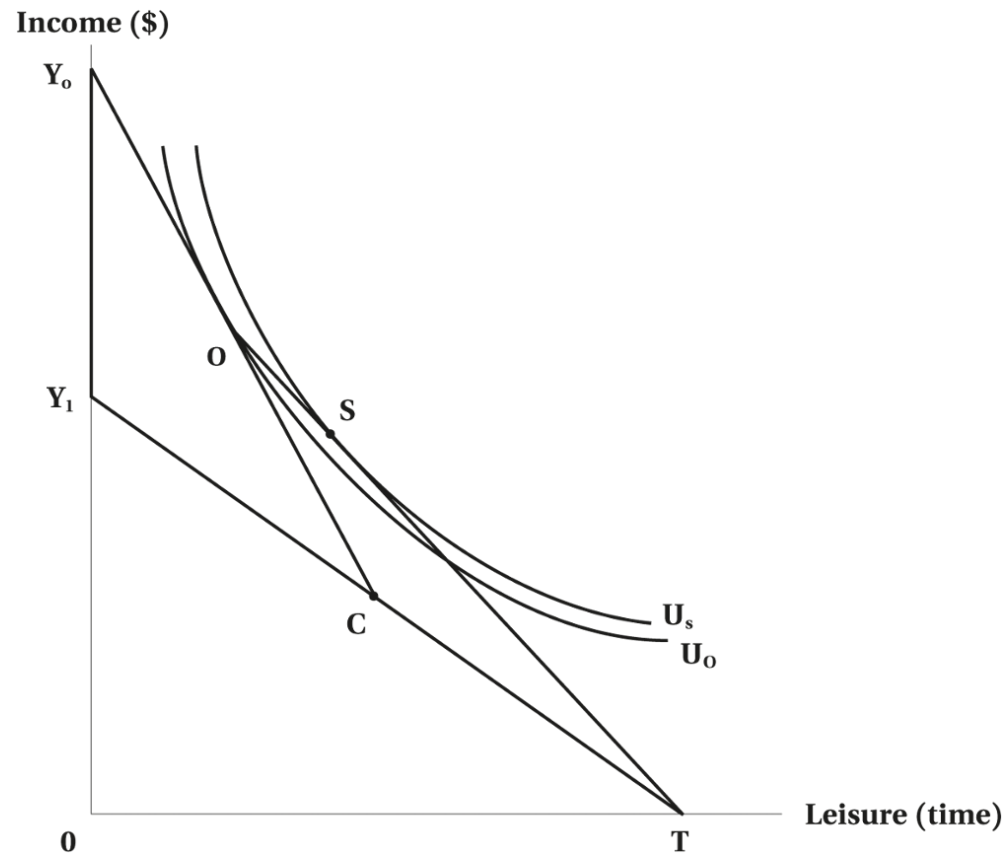


Figure 2.14 Overtime Premium versus Straight-Time Equivalent

Extensions and Applications

Choice in Working Hours

- Different groups with different preferences for work-time arrangements
- Allowing workers to work desired number of hours saves on costs
- Flex-time and compressed work week

Extensions and Applications

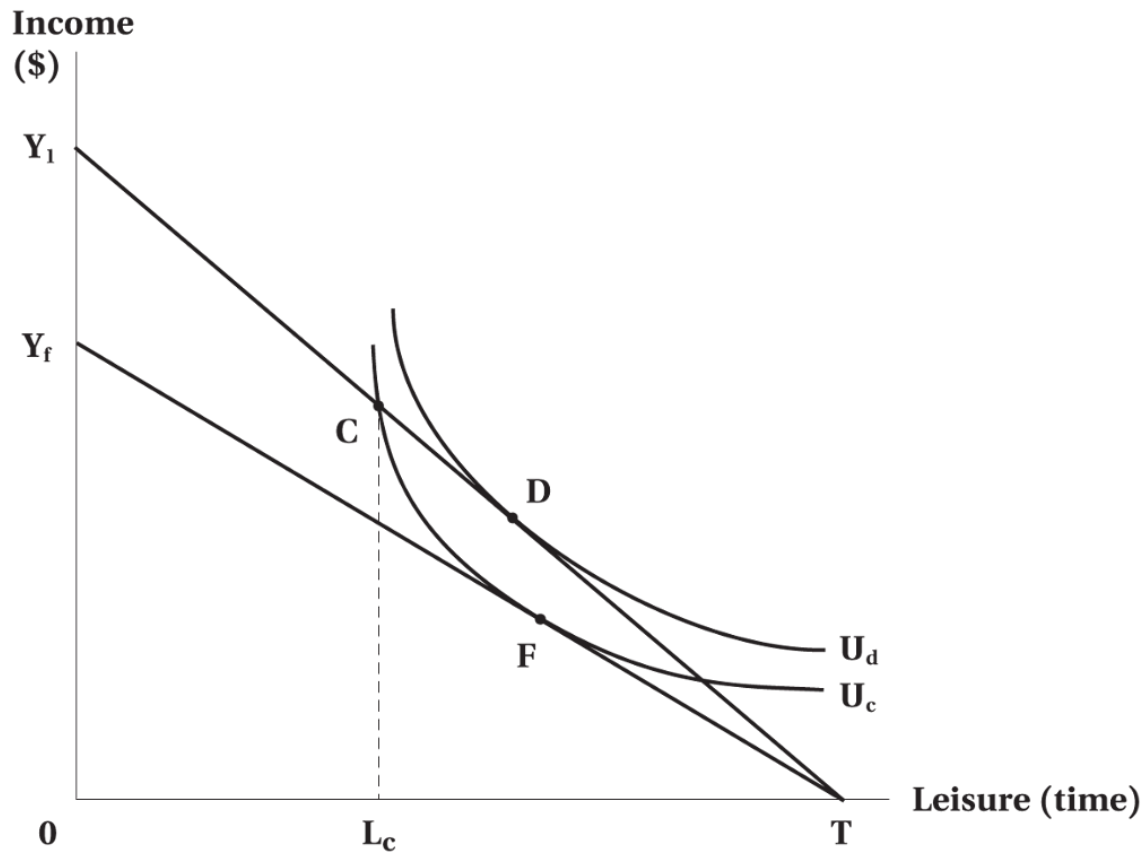


Figure 2.15 Gains from Alternative Work Schedules

Chapter Summary

- Individual attachment to the labour market is measured in two ways: first, whether an individual is working or searching for work (the participation dimension); and second, how many hours they work (the hours dimension).
- The microeconomic model of consumer choice can accommodate both dimensions of the labour supply decision. Individuals choose their preferred combination of income (consumption) and leisure (nonmarket time), as represented by their opportunity set, or potential income constraint.
- The reservation wage is the critical wage to the participation decision: for wage rates above the reservation wage, the consumer will choose to work; for wages below, the consumer will not participate.

Chapter Summary

- The labour supply model can be used to build an individual's labour supply curve. By varying the wage rate, we can trace out the individual's optimal choice of hours worked, holding all other factors constant.
- The labour supply model can be used to investigate a variety of labour market phenomena, such as the “added worker effect,” the structure of overtime wage premiums, and the willingness of individuals to accept flexibility in work hours in exchange for lower wages.